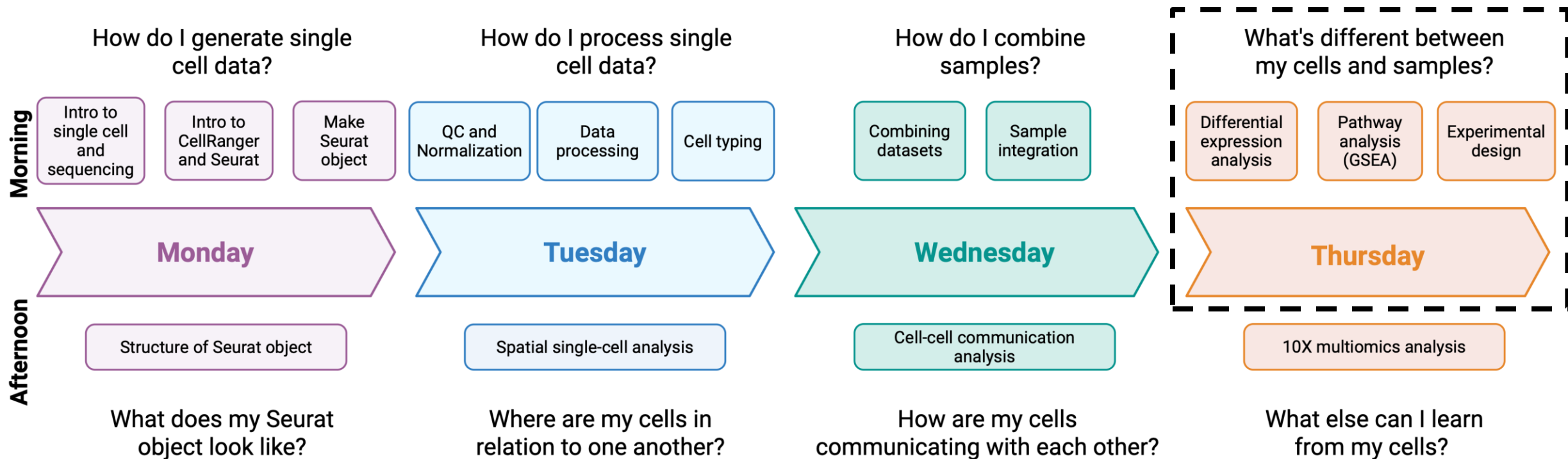


Thursday Beginner Session: Differential Expression and Gene Set Enrichment Analysis

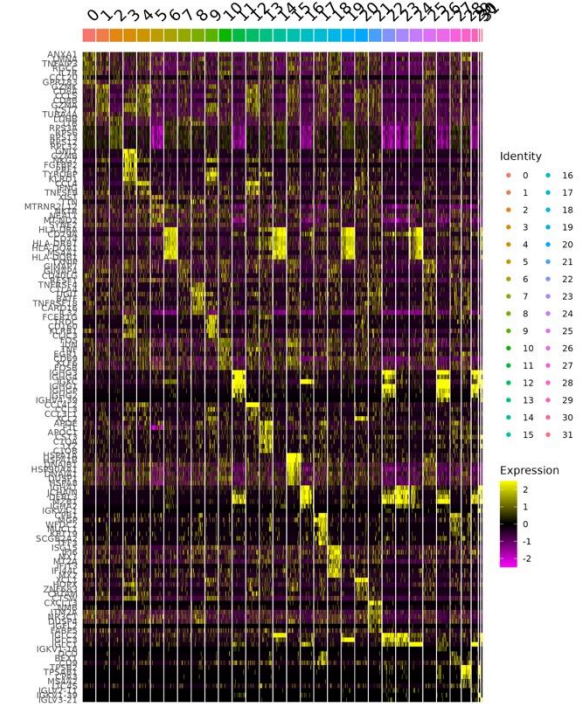
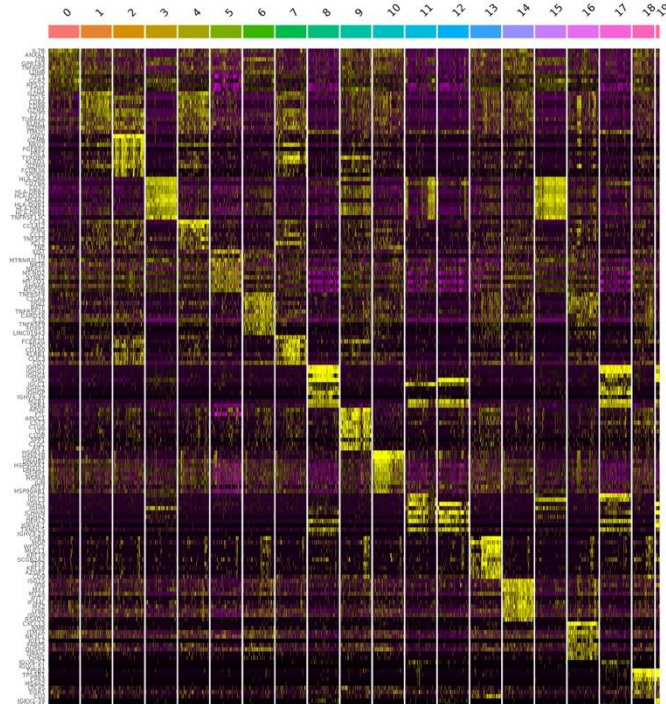
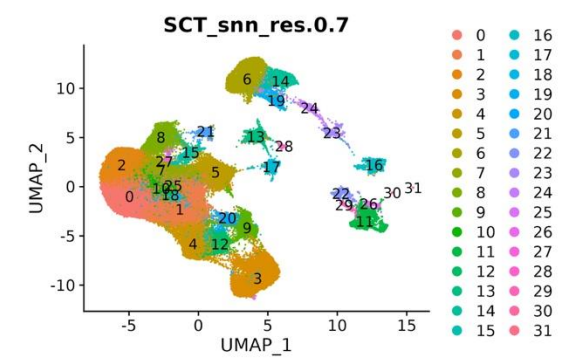
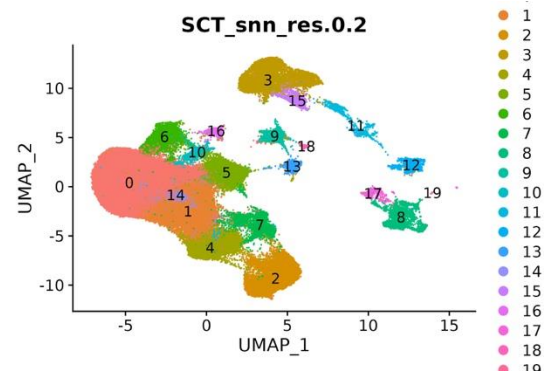


Corinne Strawser, PhD
Elizabeth Garfinkle, PhD



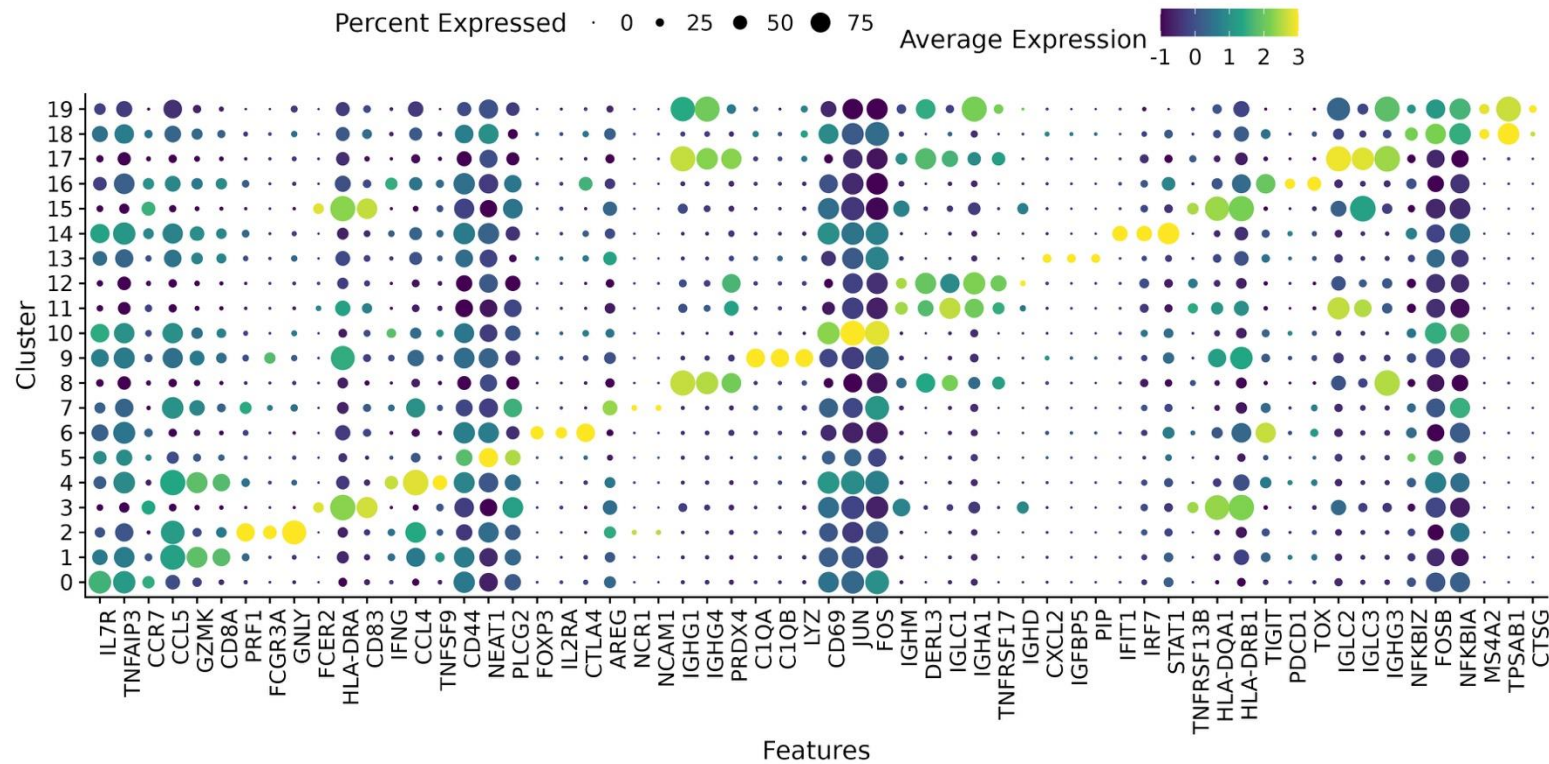
How do we use differential expression analysis in scRNA-seq?

1. Guiding clustering resolution



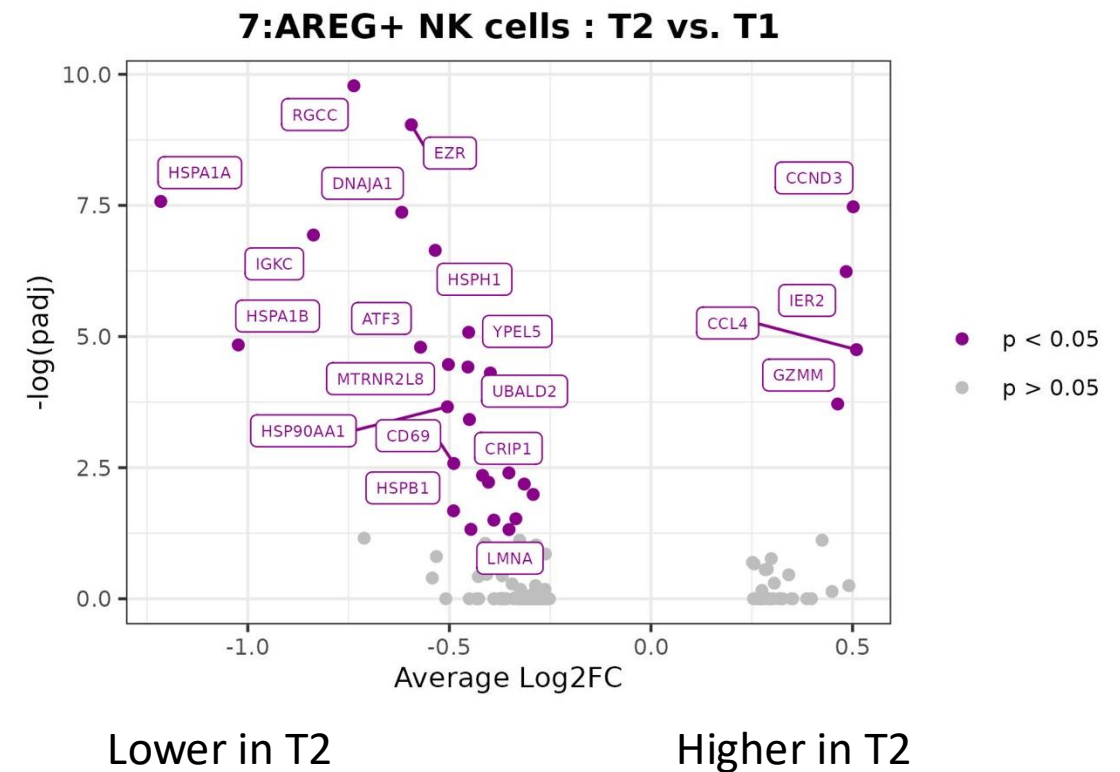
How do we use differential expression analysis in scRNA-seq?

1. Guiding clustering resolution
2. Identifying cell types



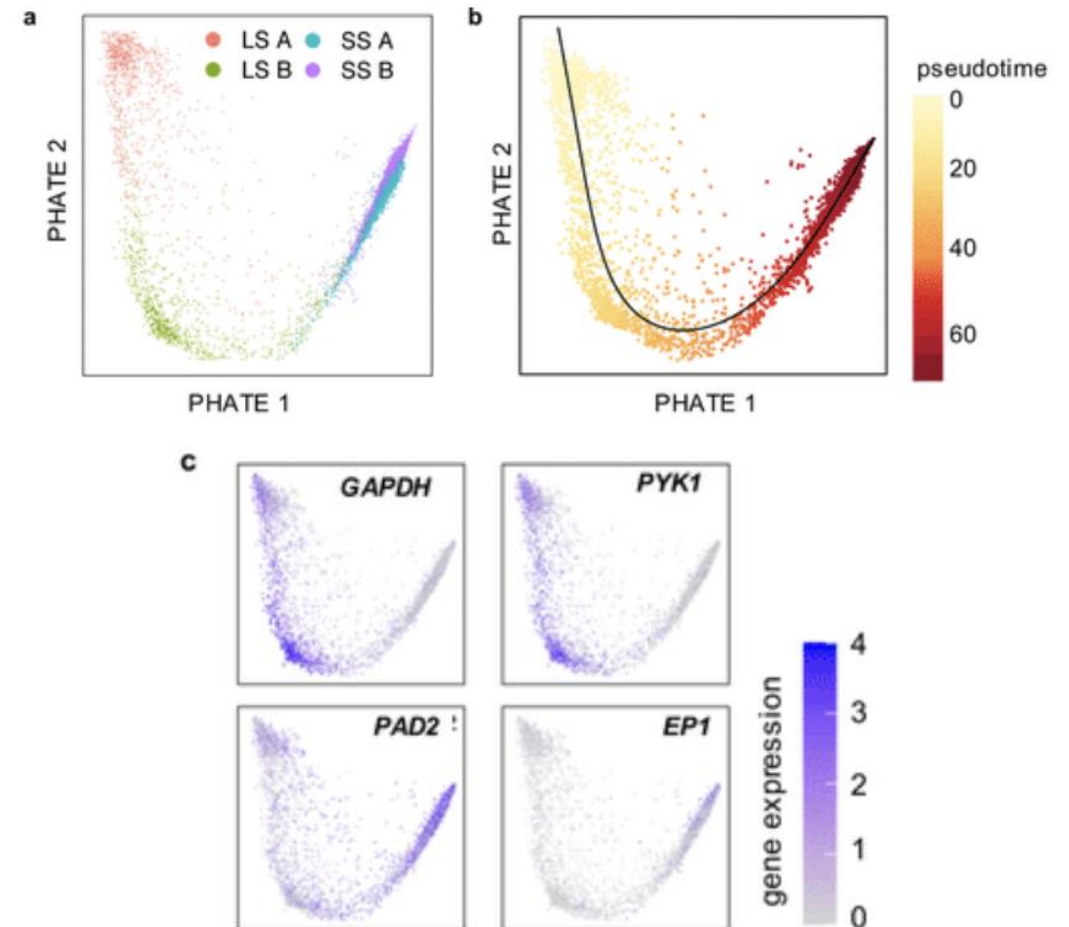
How do we use differential expression analysis in scRNA-seq?

1. Guiding clustering resolution
2. Identifying cell types
3. Identifying transcriptional differences in samples



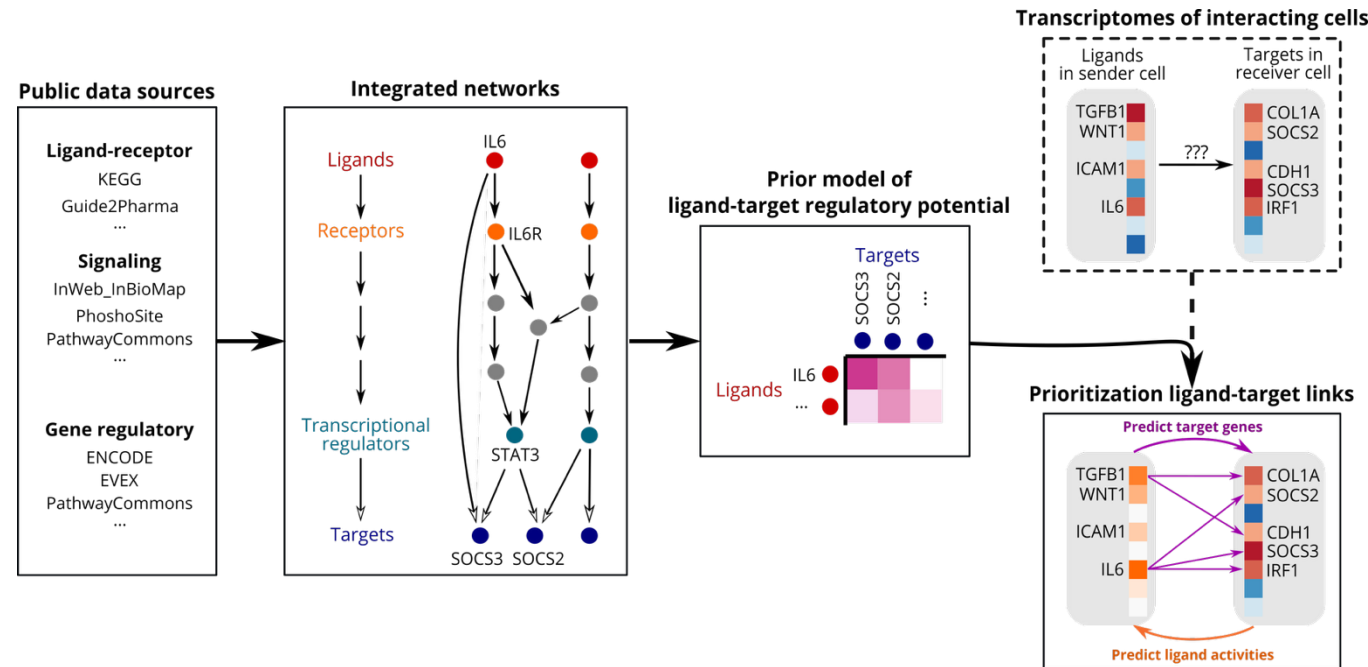
How do we use differential expression analysis in scRNA-seq?

1. Guiding clustering resolution
2. Identifying cell types
3. Identifying transcriptional differences in samples
4. Identifying transcriptional differences in pseudotime



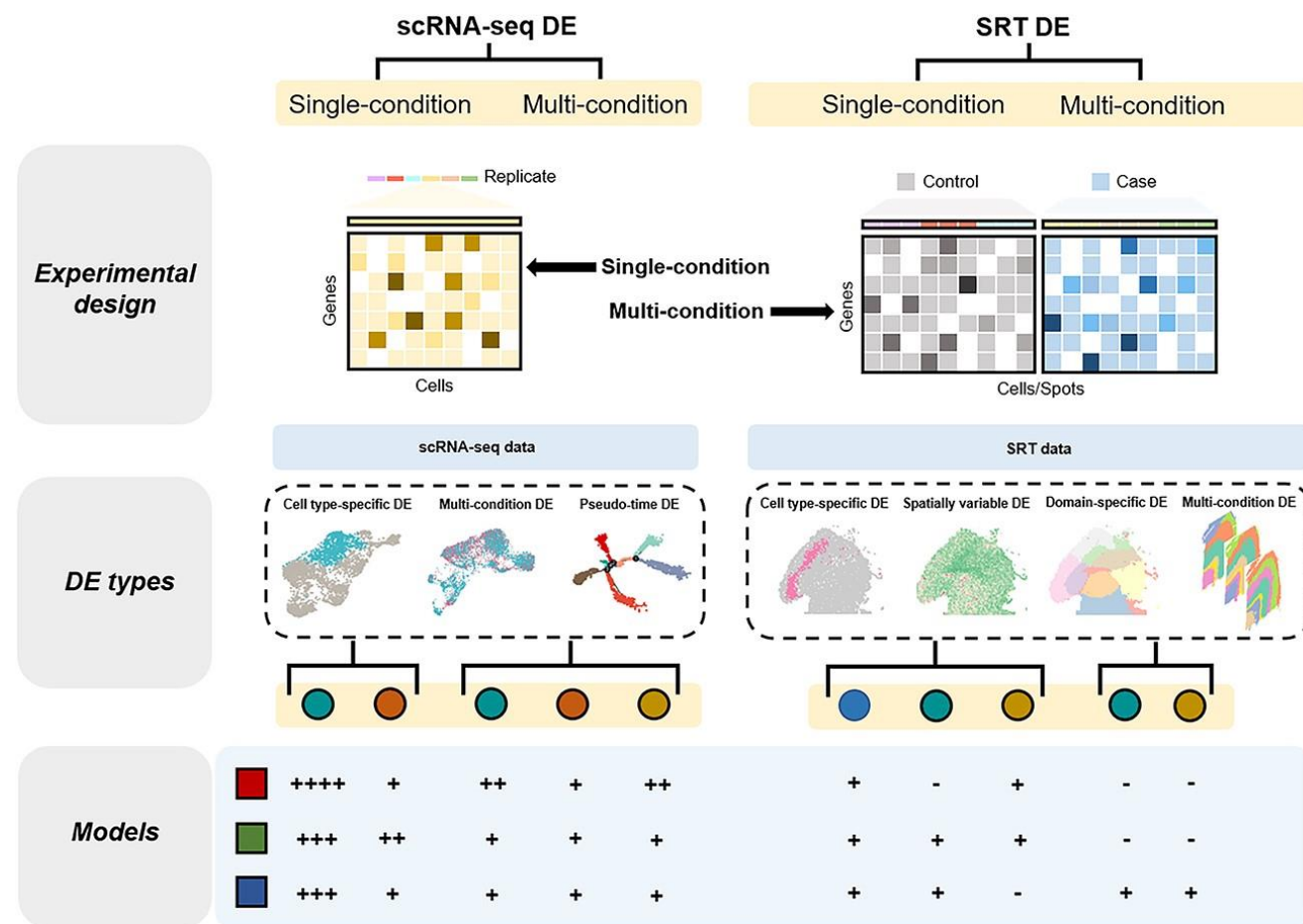
How do we use differential expression analysis in scRNA-seq?

1. Guiding clustering resolution
2. Identifying cell types
3. Identifying transcriptional differences in samples
4. Identifying transcriptional differences in pseudotime
5. Ligand-receptor analysis

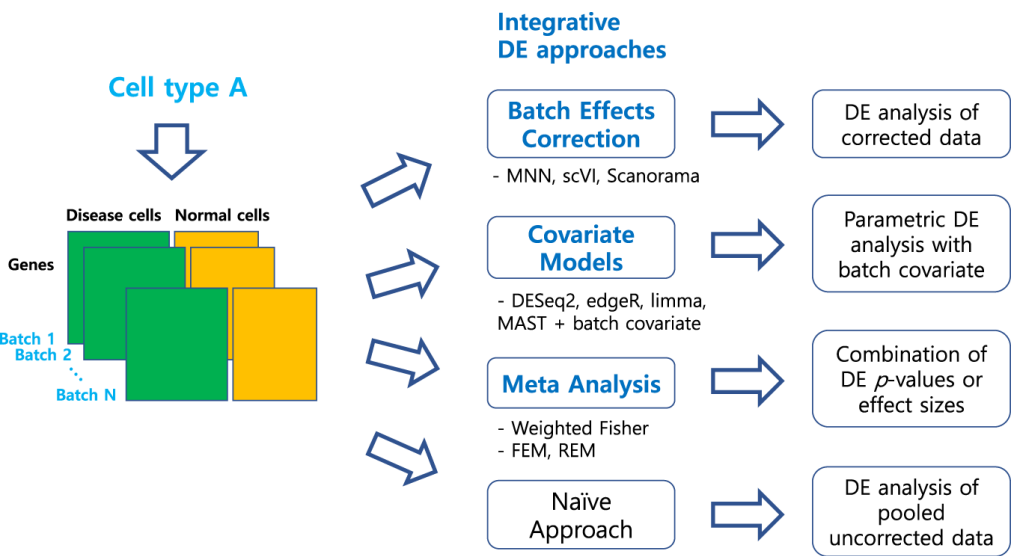


Overview of DE methods in single cell transcriptomics

- >> 130 DE tools for single cell data
- Problem #1: So many 0s!
 - Are they biological or technical?
 - Drop-out refers to a gene that is expressed but due to shallow sequencing associated with scRNA-seq, the gene is not detected resulting in a 0 read count
 - Is this missing data? should it be imputed?
- Problem #2: Some integration/batch correction processes modify the data.
 - Is this biological or technical variation?



Interaction of sparsity, sequencing depth, and batch effects



Problem #1

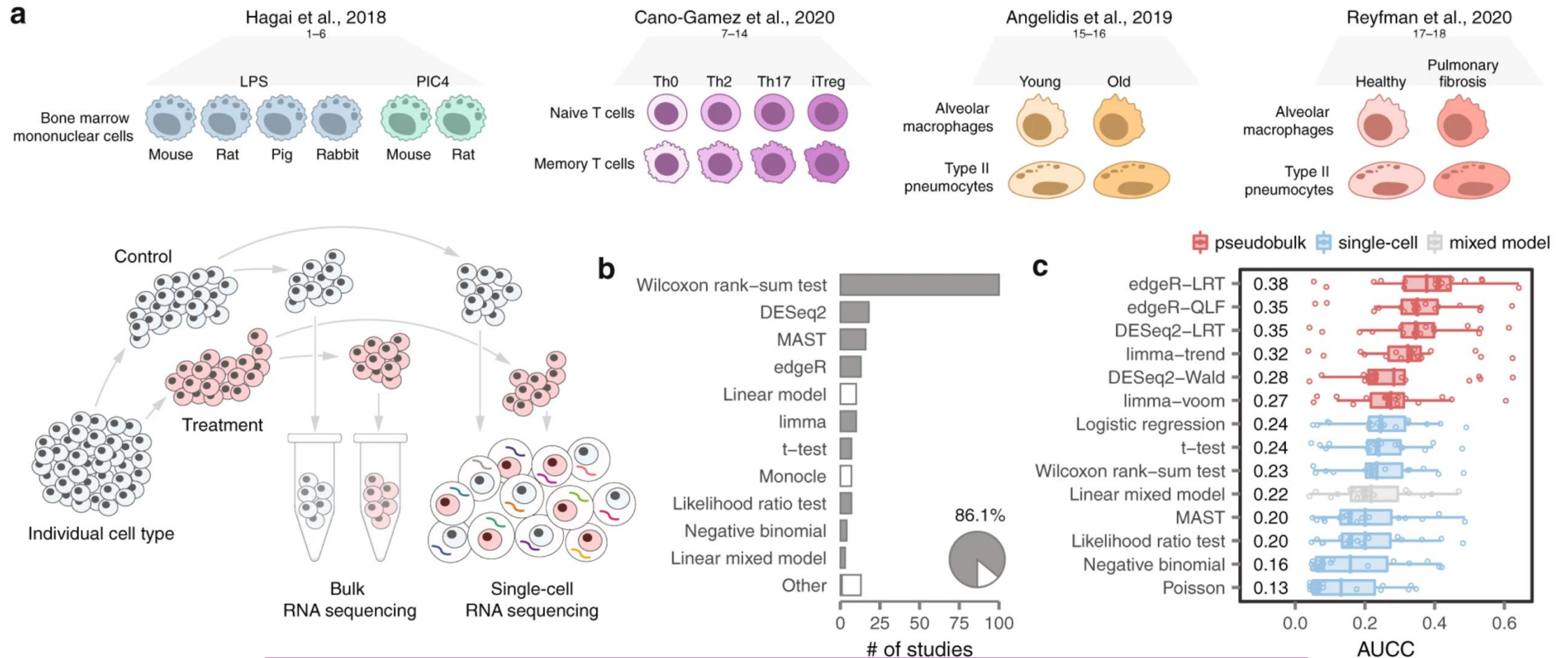
Problem #2

Table 1 | Recommended differential expression (DE) workflows for different experimental conditions

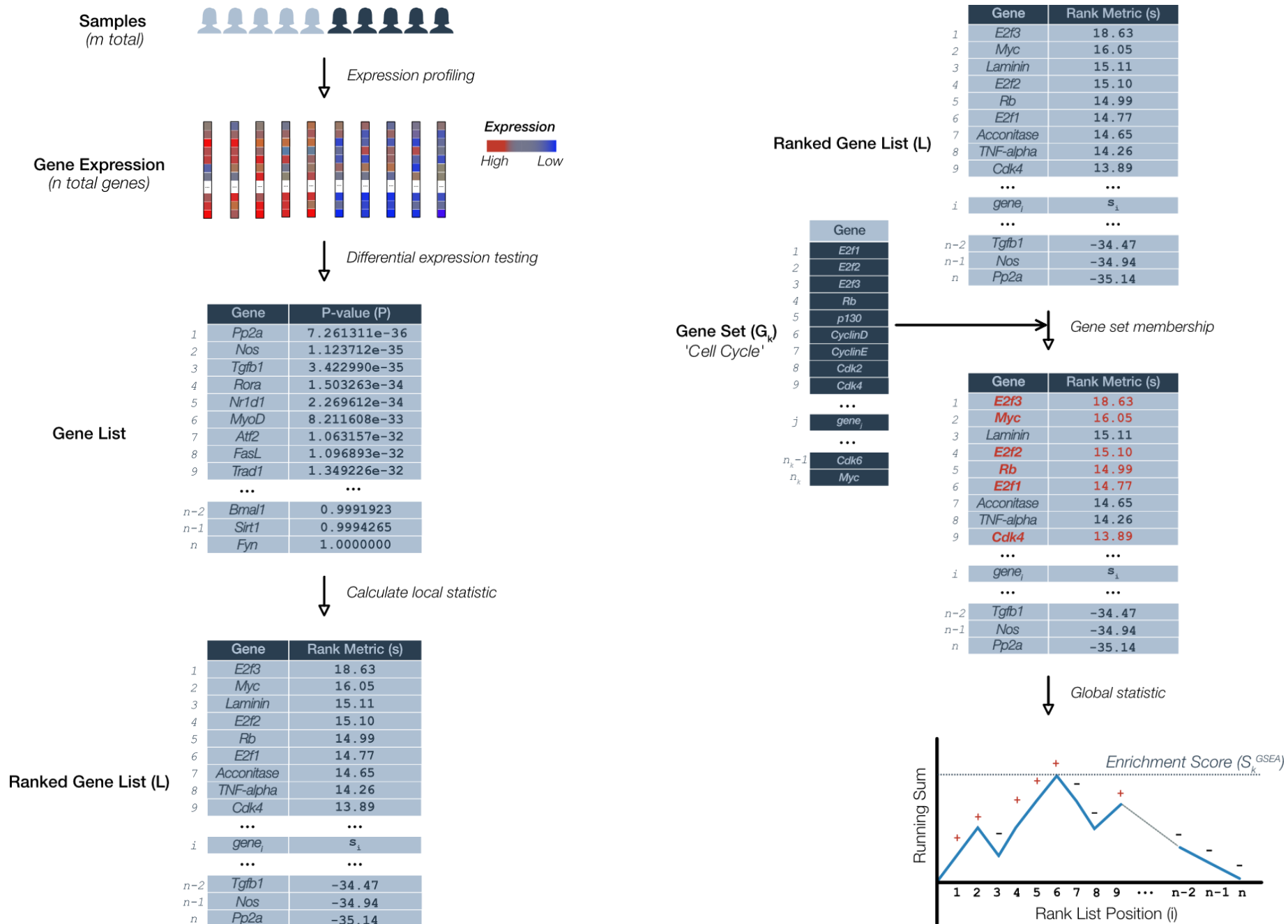
Sparsity (zero rate)	Depth*	Batch Effects	Recommended DE workflows
80%	77	Substantial	MAST_Cov, ZW_edgeR_Cov, ZW_DESeq2_Cov, scVI_limmatrend, DESeq2_FEM, limmatrend_Cov
80%	77	Small	MAST, ZW_edgeR, ZW_DESeq2, Pseudobulk_limma, DESeq2_FEM, limmatrend_Cov
80%	10	Substantial	DESeq2_Cov, limmatrend_Cov, DESeq2_wFisher, LogN_FEM, MAST_Cov, Raw_Wilcox
80%	10	Small	DESeq2, limmatrend_Cov, LogN_FEM, Pseudobulk_edgeR, Pseudobulk_limma, Raw_Wilcox
80%	4	Substantial	LogN_FEM, limmatrend, Raw_Wilcox, RISC_QP
80%	4	Small	LogN_FEM, limmatrend, Raw_Wilcox, RISC_QP
40%	77	Substantial	MAST_Cov, ZW_edgeR_Cov, ZW_DESeq2_Cov, limma_BEC_Wilcox, Scanorama_limmatrend, logN_FEM
40%	77	Small	Pseudobulk_limma, Raw_Wilcox, Pseudobulk_edgeR, DESeq2, LogN_FEM
40%	4	Substantial	limmavoom_Cov, limmatrend_Cov, ZW_edgeR_Cov, ZW_DESeq2_Cov, logN_FEM, limma_BEC_Wilcox
40%	4	Small	Pseudobulk_limma, limmvoom, limmatrend, Pseudobulk_edgeR, Raw_Wilcox, edgeR

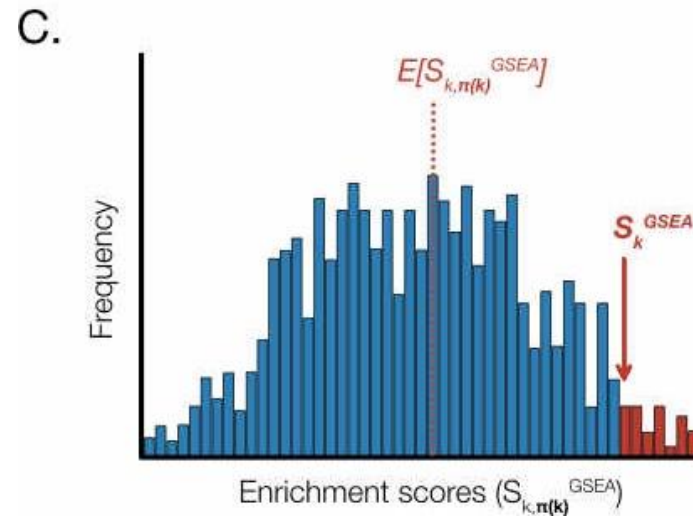
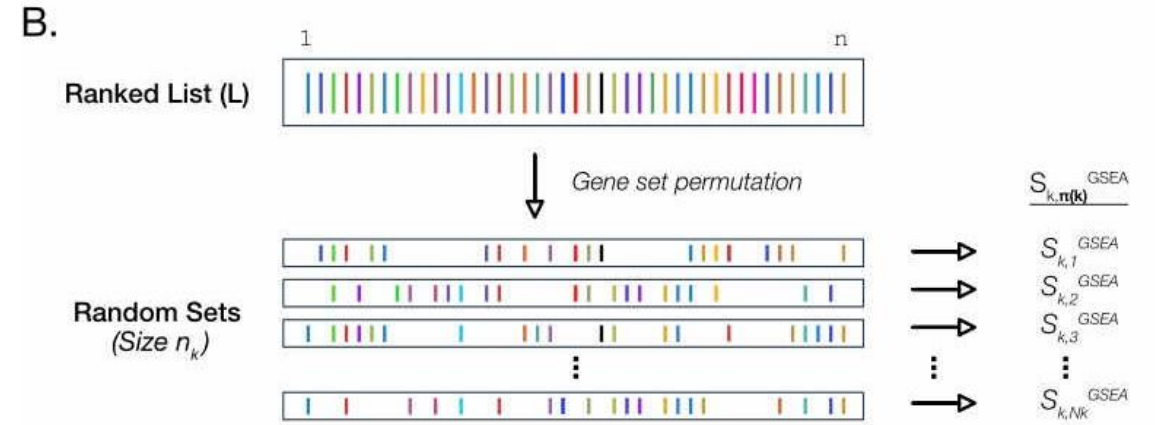
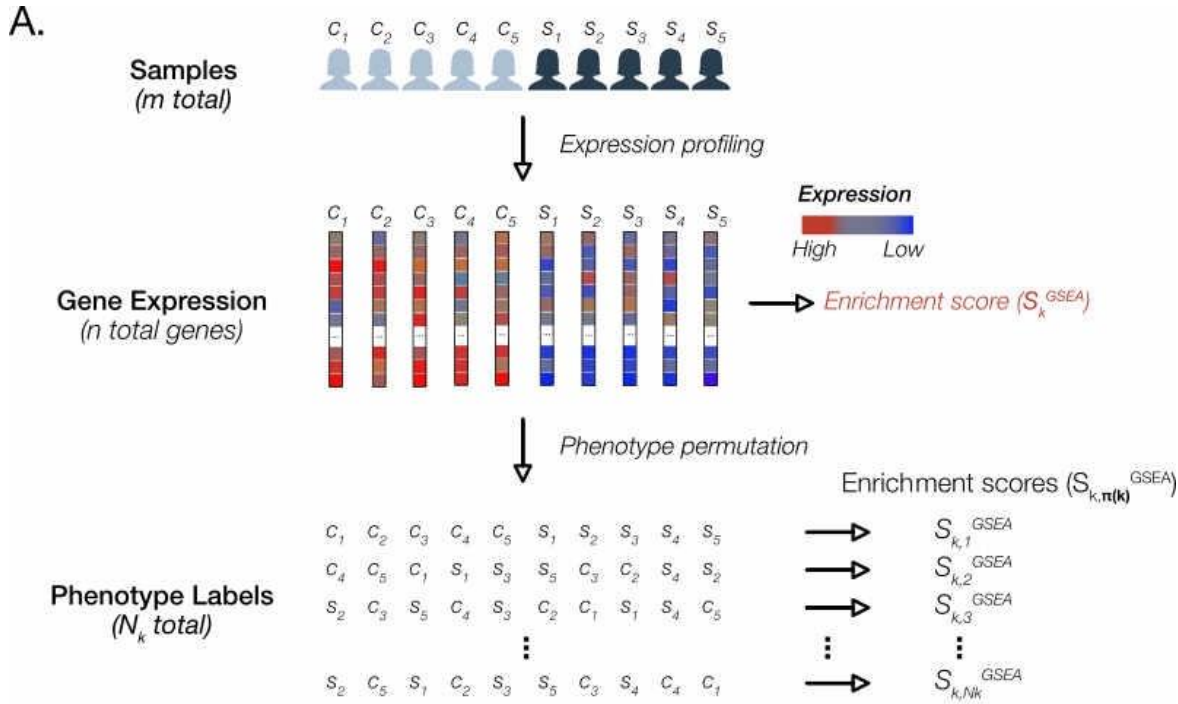
In 40% sparsity cases, recommended methods were selected based on simulation results only.
*Average nonzero count in each cell after filtering sparsely expressed genes (zero rate > 0.95).

Best practices for single-cell differential expression analysis are still being developed



NOTE: Pseudobulk methods should have at least 3 samples per group

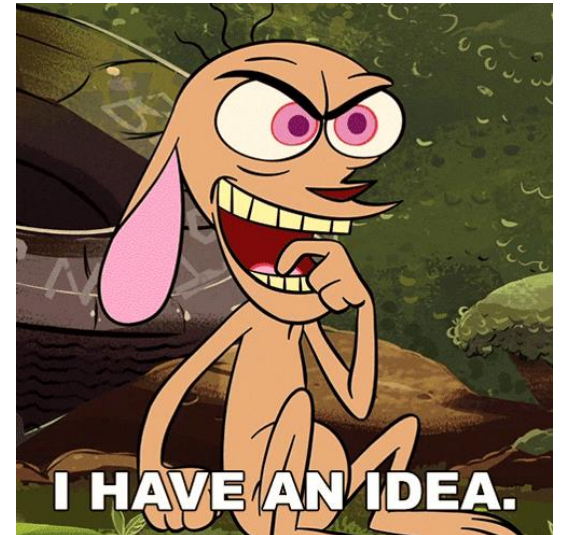




- + Normalize Enrichment Scores (NES) to Gene Set Size
- + Multiple Test Corrections

Experimental Design Considerations

1. Do I really need to do scRNA-seq?
2. What type of protocol do I need to use?
3. How much money do I have?
4. How do I mitigate batch effects?



Do I really need to do scRNA-seq?

- single cell type of interest - probably not
- landscape of cell types in a sample – probably
- Make sure you think through your specific single cell or spatial question
- TIP: Data is often hypothesis generating - should be validated with secondary assay (ex: qPCR)



What type of protocol do I need to use?

- species of interest (reference species vs. unusual organism)
- transcript capture (poly-A 3' GEX, poly-A 5' GEX, probe-based)
- number of samples (multiplex vs. single plex)
- frequency of cell types of interest (FACS sorting, plate-based)
- frequency of transcripts of interest (droplet-based, plate-based)
- ability to identify cell types of interest (protein + GEX assays)
- input cell type/quality (fresh vs frozen vs fixed vs FFPE vs OCT)
- “Multiomics” – GEX paired with ATAC/TCR+BCR/Spatial/etc.



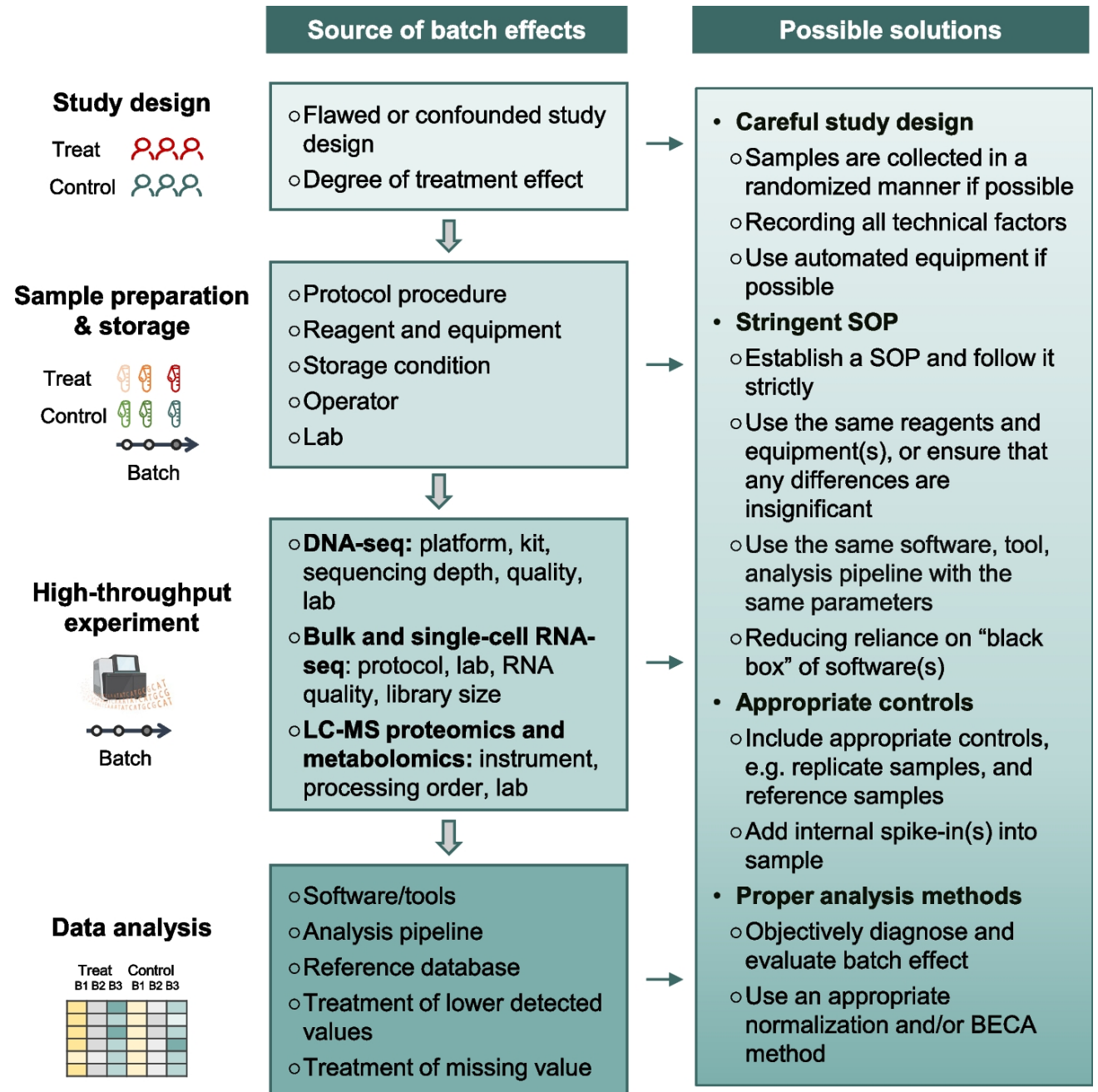
How much money do I have?

- capital investment (10X Chromium vs. PIP-seq vs. plate-based)
- number of samples
- $n=1$ per condition still acceptable, but $n=3$ or more allows for pseudobulking and DESeq2 differential expression
- sequencing requirements



How do I mitigate batch effects?

TIP: don't separate samples into different lanes/sequencing runs if you want to compare them!





PRO TIP: NCH Genomic Services Laboratory (GSL) has single cell core to help with planning and executing your experiments. Please reach out if you're interested!

Contact: Dr. Katie Miller, GSL Director
(Katherine.Miller@nationwidechildrens.org)



- **EXPERT TIP:** The scRGOT group meets monthly and is here to help and share ideas! Come join us!

Get on the mailing list!

Contact: Dr. Katie Miller

(Katherine.Miller@nationwidechildrens.org)



- We also have a weekly bioinformatics meeting on Thursdays at 1PM!

Contact: Dr. Matthew Cannon

(Matthew.Cannon@nationwidechildrens.org)

Best practices for single-cell analysis across modalities

Lukas Heumos ^{1,2,3,28}, **Anna C. Schaar** ^{1,4,5,28}, **Christopher Lance** ^{1,6}, **Anastasia Litinetskaya**^{1,4}, **Felix Drost** ^{1,3},
Luke Zappia ^{1,4}, **Malte D. Lücken** ^{1,7}, **Daniel C. Strobl**^{1,3,8,9}, **Juan Henao**¹, **Fabiola Curion** ^{1,4}, **Single-cell Best Practices Consortium***, **Herbert B. Schiller**² & **Fabian J. Theis** ^{1,3,4,5} 



Search

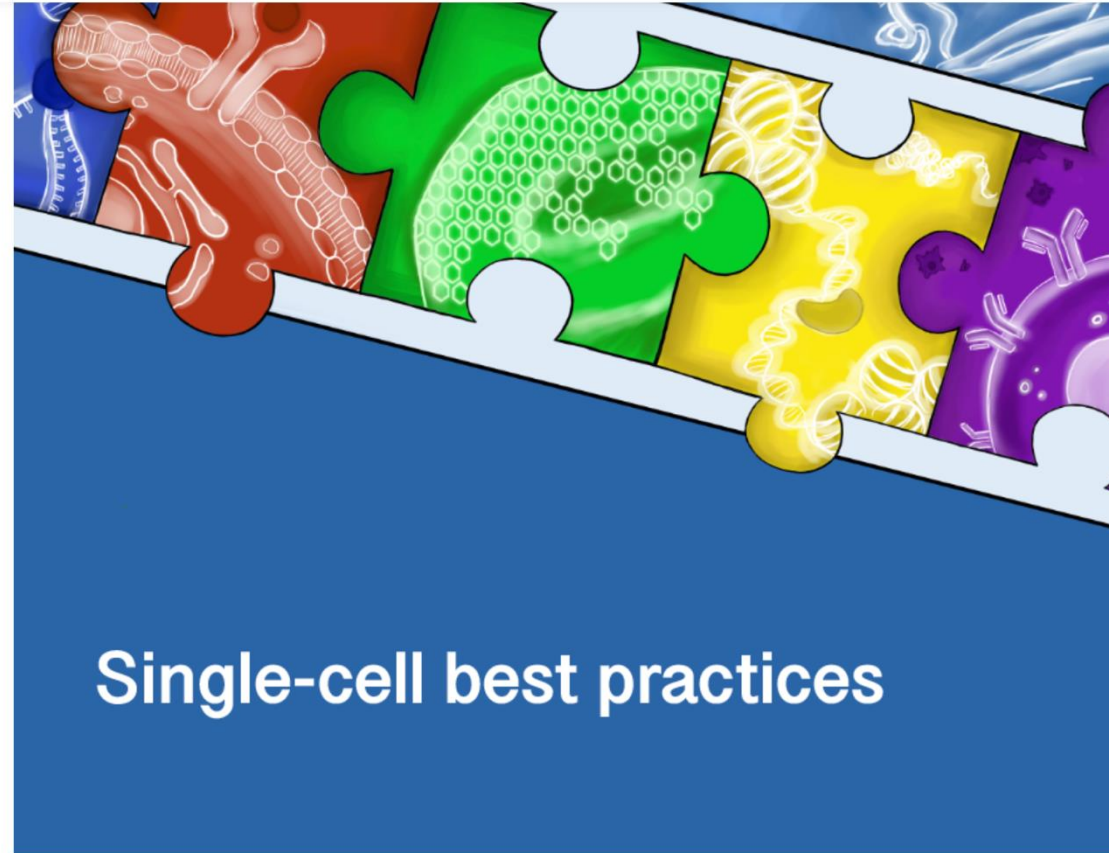
* + K

Introduction

1. Prior art
2. Single-cell RNA sequencing
3. Raw data processing
4. Analysis frameworks and tools
5. Interoperability

Preprocessing and visualization

6. Quality Control
7. Normalization
8. Feature selection
9. Dimensionality Reduction

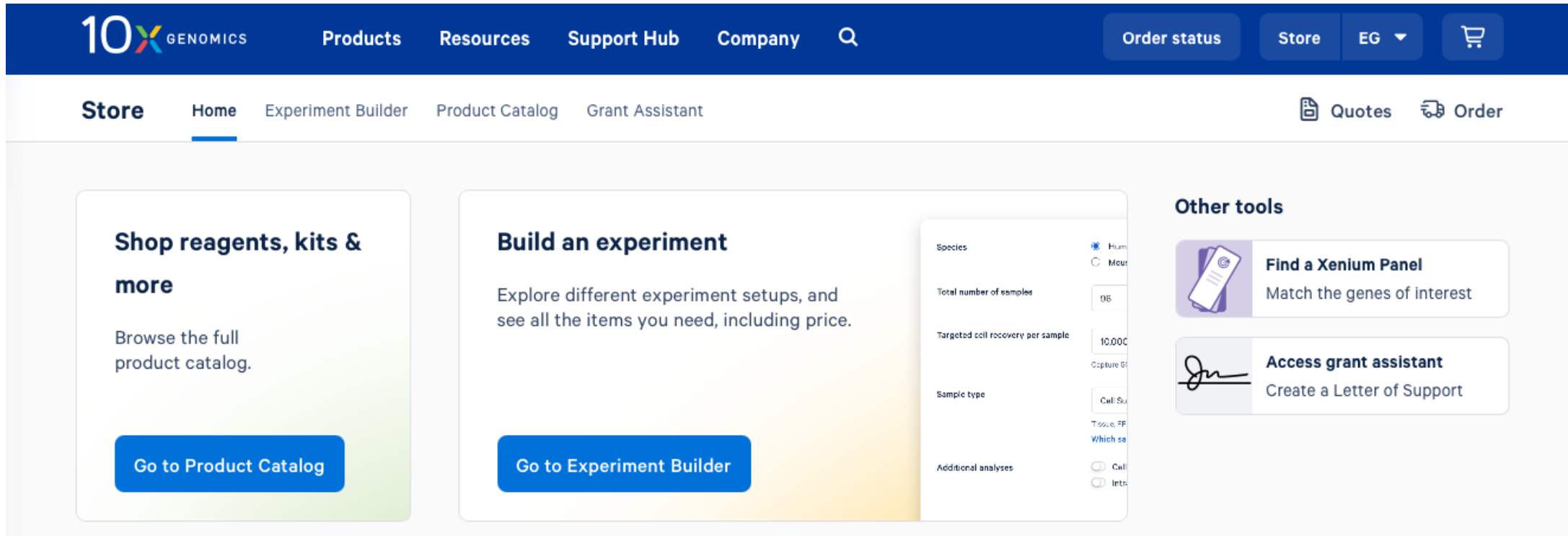


Single-cell best practices

Contents

- Introduction
- What this book covers
- What this book does not cover
- Who should read this book
- Structure of the book
- Prerequisites
- Peer-review
- Citation
- Contributing
- Alternative formats
- Contact us
- License
- References

10x Genomics Can Help! <https://www.10xgenomics.com/store>



The screenshot shows the 10x Genomics Store homepage. The top navigation bar is dark blue with the 10x Genomics logo on the left and links for Products, Resources, Support Hub, and Company in the center. On the right of the top bar are buttons for Order status, Store, EG (with a dropdown arrow), and a shopping cart icon. Below the top bar is a secondary navigation bar with links for Store, Home (underlined), Experiment Builder, Product Catalog, and Grant Assistant. On the right of this bar are icons for Quotes and Order. The main content area features three large cards. The first card, 'Shop reagents, kits & more', has a green gradient and a 'Go to Product Catalog' button. The second card, 'Build an experiment', has a yellow gradient and a 'Go to Experiment Builder' button. To the right of this card is a preview of the Experiment Builder interface, showing fields for Species (Human/Mouse), Total number of samples (96), Targeted cell recovery per sample (10,000), Sample type (Cell Suspension/Tissue/Purified), and Additional analyses (Cell Intracellular). The third card, 'Other tools', contains two sub-sections: 'Find a Xenium Panel' with a button to 'Match the genes of interest', and 'Access grant assistant' with a button to 'Create a Letter of Support'.

10x GENOMICS Products Resources Support Hub Company [Order status](#) [Store](#) [EG](#) [Shopping Cart](#)

[Store](#) [Home](#) [Experiment Builder](#) [Product Catalog](#) [Grant Assistant](#) [Quotes](#) [Order](#)

Shop reagents, kits & more

Browse the full product catalog.

[Go to Product Catalog](#)

Build an experiment

Explore different experiment setups, and see all the items you need, including price.

[Go to Experiment Builder](#)

Species: ☒ Human ☐ Mouse

Total number of samples:

Targeted cell recovery per sample:

Sample type:

Additional analyses:

Other tools

**Find a Xenium Panel**

Match the genes of interest

**Access grant assistant**

Create a Letter of Support

Experiment Builder

10x Genomics Can Help! <https://www.10xgenomics.com/store>

Which assay are you using for this experiment?

Assay

Chromium Single Cell

Flex Gene Expression

Version

v2 - GEM-X / Apex

Latest

Continue



Our most sensitive whole transcriptome gene expression, with flexibility to fix, batch, and run samples on your schedule

- ✓ All sample types, including FFPE
- ✓ Probe-based gene expression, with optional Cell Surface Protein
- ✓ Highly scalable, with lowest-cost per cell

[Learn more about Flex](#)

[Compare Flex with alternatives](#)

Experiment Design: Visualize each step of your run and see your samples progress into final libraries on the right. For pricing, go to the [Pricing](#) tab.

Run pilot and smaller scale experiments

Reset Fields

Species

- ☒ Human
☐ Mouse
☐ Rat

Total number of samples

16

Ideal for 1 to 32 samples.

Targeted cell recovery per sample

20,000

Sample type

Cell Suspension

Tissue, FFPE, whole blood, and more.

[Which sample type should I choose?](#)

Multiplexing

- ☐ Singleplex
☒ Multiplex samples

Additional analyses

- ☐ Cell Surface Protein
☐ Intracellular Protein

Pricing

Experiment design

If you are new to 10x Genomics technology, view our [Terminology Guide](#) first.

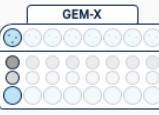
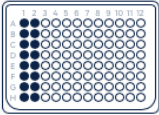
16 samples barcoded in 16 wells with 16 sample barcodes

1 well = 1 sample = 1 sample barcode

1 multiplexed pool (16 samples) created

1 channel used in 1 chip (1 Gel Bead reaction)

4 final sequencing libraries created



Add Experiment to Cart

We've verified your affiliation with **Nationwide Children's Hospital - Institute for Genomic Medicine.**

Questions? We're here to help.
Contact Joseph Ipe, your 10x Account Manager at joseph.ipe@10xgenomics.com